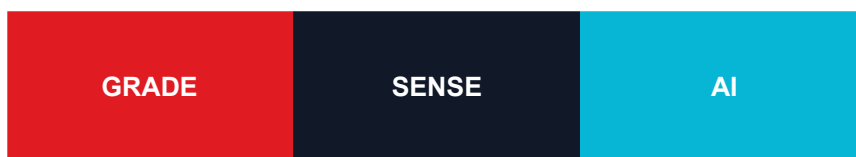


GradeSense Technical Documentation

Architecture, intelligence workflows, explainability, deployment, and results



Explainable predictions, constraint-aware recommendations, and real-time operations for safer, faster paper grade transitions.

Prepared from the validated GradeSense v0.2.0 source repository and live application.

Executive Summary

GradeSense | Honeywell Hackathon Submission

GradeSenseAI is an operator-facing industrial intelligence platform for paper-mill grade transitions. It combines snapshot prediction, sequential basis-weight forecasting, explainable AI, safe intervention simulation, persistent recommendations, alerts, and operational governance in one deployable system.

The platform is advisory by design: every action is constraint-checked, model-simulated, ranked by expected improvement, explained in operational language, and preserved for later effectiveness analysis.

- Decision support spans live sensing, inference, recommendation, operator action, and outcome learning.
- REST and WebSocket interfaces support both deliberate workflows and continuous streaming.
- Immutable model artifacts, audit logs, health checks, and registry controls support production readiness.

Problem Statement

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Paper grade transitions are multivariable, time-dependent processes. Operators must stabilize basis weight and quality while machine speed, moisture, stock flow, steam pressure, temperature, pulp consistency, and grade-specific targets change together.

The objective is to reduce off-spec production and stabilization time without replacing operator authority.

Existing Challenges

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- Delayed detection: conventional thresholds react after deviation is visible.
- Fragmented context: sensor trends, model confidence, and historical outcomes live in separate views.
- Unsafe optimization: an apparently beneficial set-point change can violate process constraints.
- Low trust: black-box predictions do not explain why risk changed or what to do next.
- Weak learning loop: accepted recommendations and observed outcomes are rarely measured together.

Proposed Solution

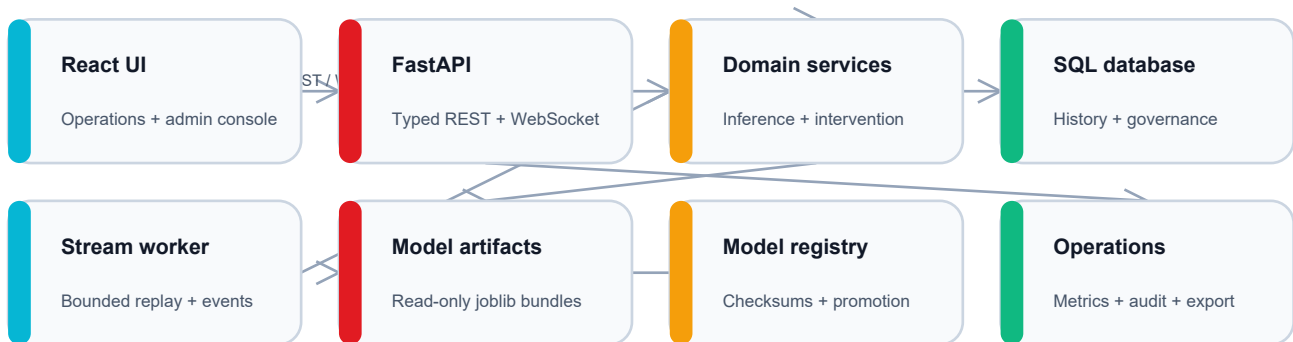
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GradeSense fuses typed FastAPI services, a React operations console, persisted operational history, and two model families. Snapshot intelligence estimates current transition quality, off-spec risk, and stabilization; sequential forecasting projects basis-weight trajectory and detects specification crossing.

- Generate only feasible intervention candidates.
- Simulate each candidate with the active forecast model.
- Rank improvements deterministically by crossing risk, peak deviation, and stabilization.
- Expose feature-level reasons and confidence instead of opaque scores.
- Stream predictions, alerts, drift, and recommendations to the dashboard.

System Architecture

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ML Pipeline

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The snapshot pipeline validates and preprocesses a transition state, resolves the active model, produces quality/risk/stabilization outputs, and attaches feature-level explanations. The sequential pipeline enforces transition boundaries, builds lag/rolling/rate features, executes direct-horizon regressors, estimates residual intervals, and monitors the $\pm 2.5\%$ specification band.

- Direct-horizon forecasting avoids recursive error accumulation.
- Windows never cross transition IDs, protecting temporal validity.
- Validation residuals become prediction intervals.
- Model and feature-schema checksums make inference reproducible.

Recommendation Engine

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The intervention engine creates one-to-four-variable candidate adjustments from operationally meaningful ranges. A constraint engine rejects unsafe combinations before inference. Surviving candidates are simulated against a persisted baseline and ranked only when they improve the forecast.

- Candidate generation: single-variable and coordinated multivariable actions.
- Safety gate: bounds, change magnitude, and incompatible combinations.
- Benefit gate: crossing avoidance, lower peak deviation, or faster stabilization.
- Lifecycle: proposed, accepted, rejected, modified, delayed, applied, evaluated.

Explainable AI

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GradeSense turns model evidence into operator-facing reasoning. Snapshot explanations surface the most influential variables and their direction. Forecast recommendations state the proposed set-point changes, why they

are safe, which risk measure improves, and the expected operational effect.

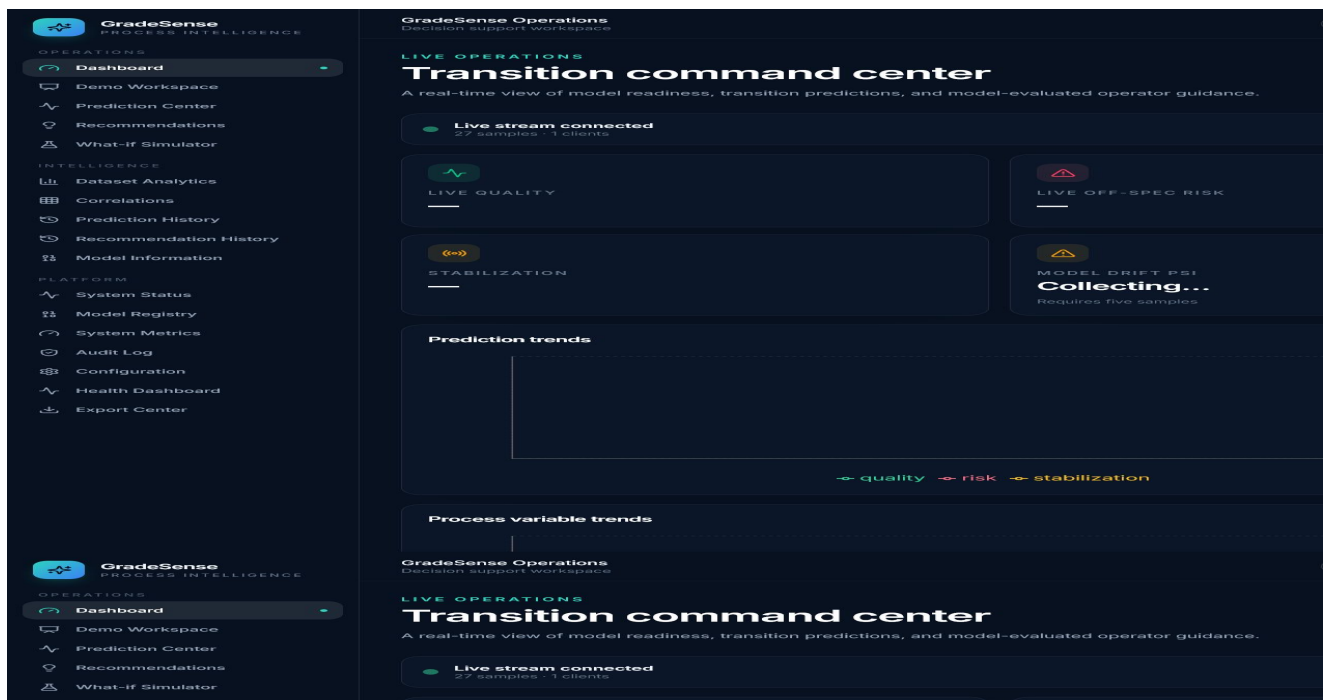
Confidence, model version, data recency, and drift are visible so the operator can judge when the model should carry less weight.

Dashboard

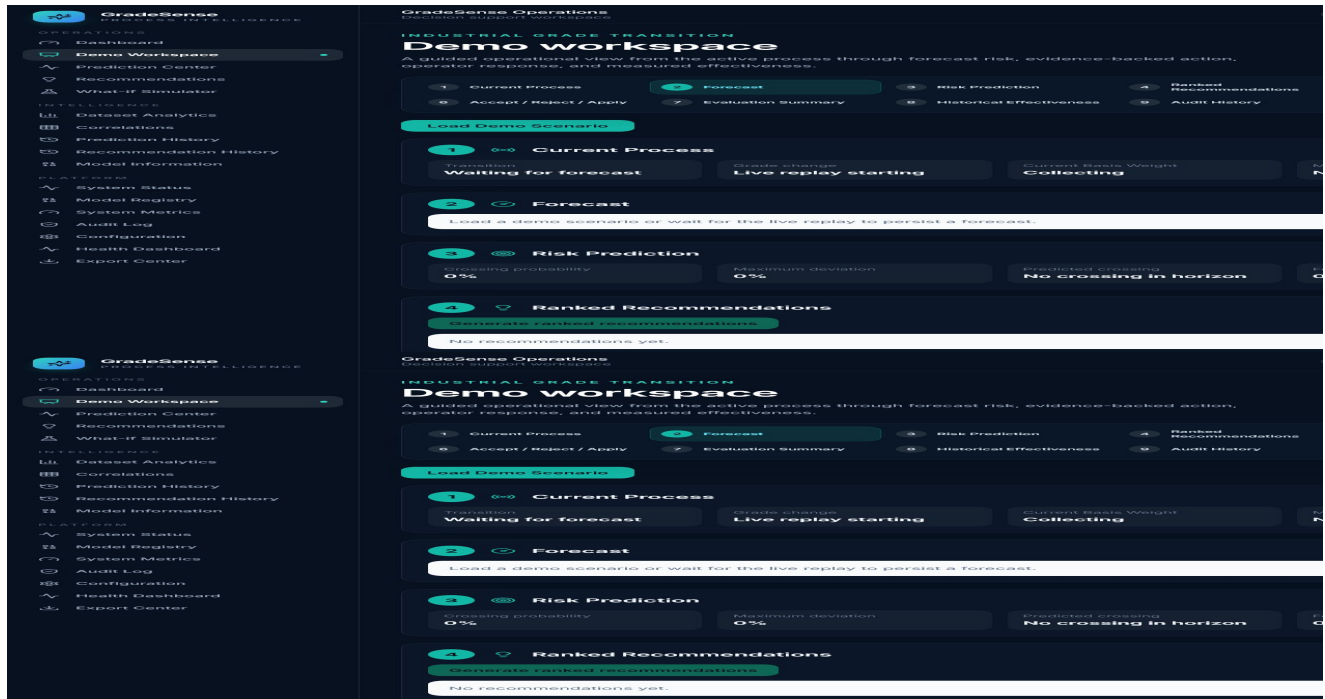
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The responsive dashboard provides a transition command center, live charts, rolling metrics, alerts, prediction and recommendation workflows, analytics, searchable history, model registry, health, audit, configuration, and export controls.

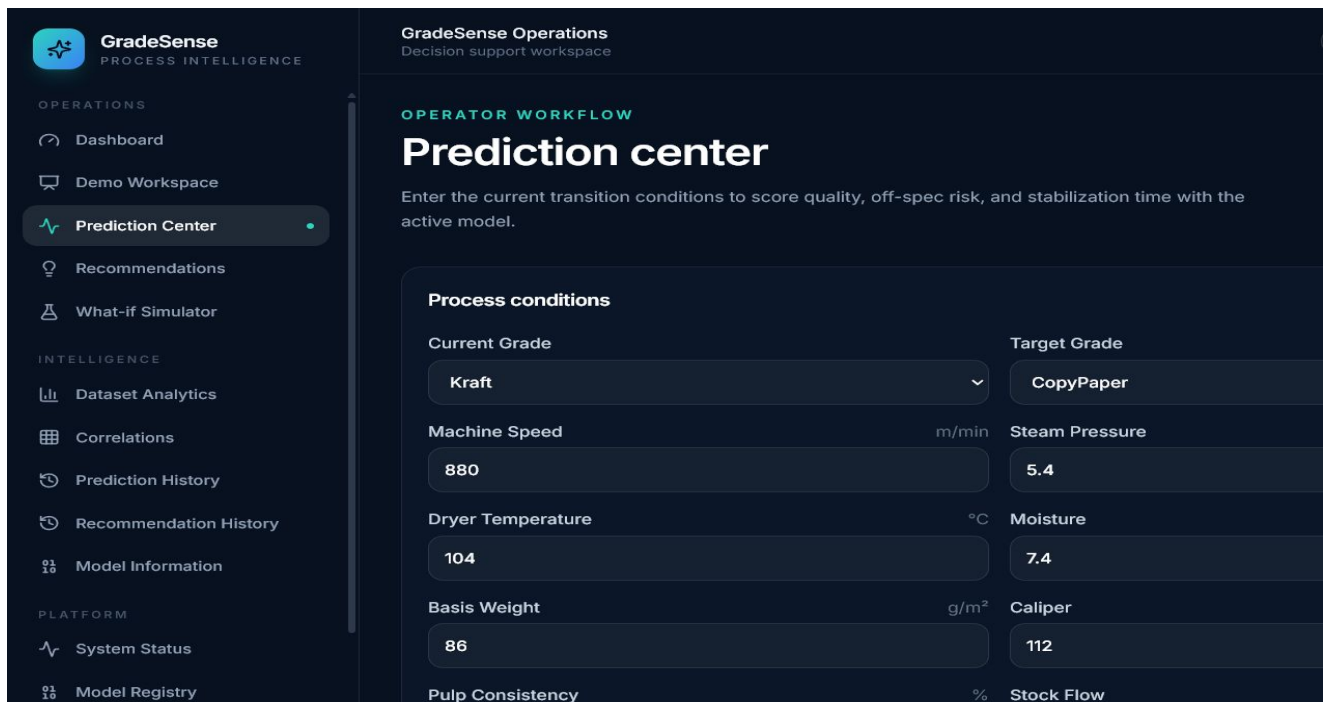
Home



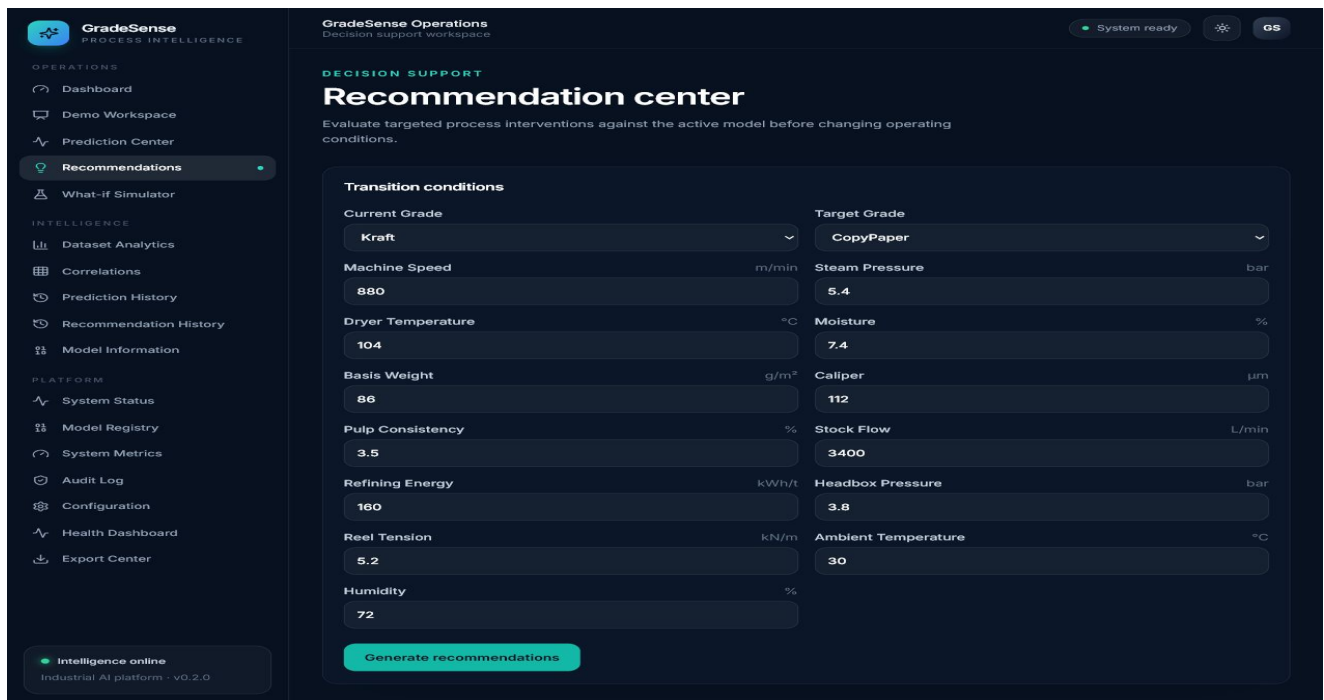
Live Dashboard



Prediction

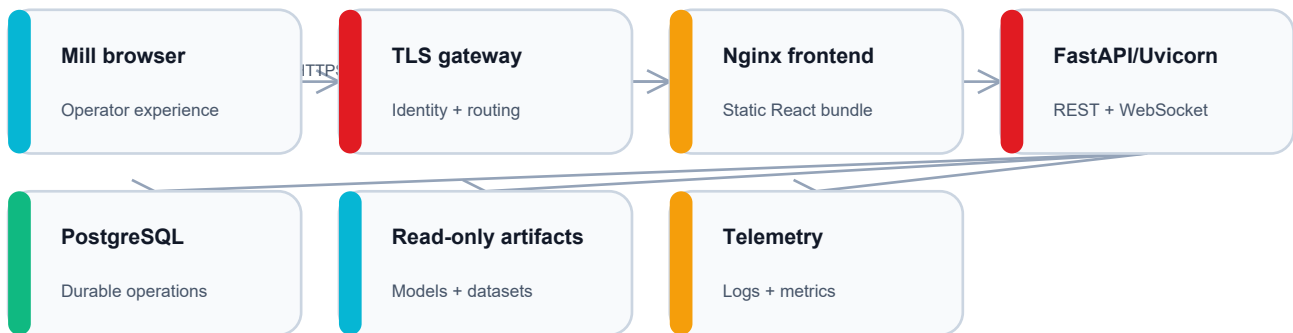


Recommendation



Deployment Architecture

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API Overview

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- System: identity, health, version, OpenAPI.
- Intelligence: predict, recommend, correlations, dataset, model, history.
- Forecasting: trajectory, history, simulation, relationship discovery.
- Interventions: ranked actions, decisions, outcomes, effectiveness.
- Real time: alerts, feedback, stream state, rolling/live metrics, drift, WebSockets.
- Administration: registry, configuration, metrics, health, audit, exports.

Technologies

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- Backend: Python, FastAPI, Pydantic, SQLAlchemy, Alembic, Uvicorn.
- ML: scikit-learn-compatible joblib artifacts, feature engineering, direct-horizon regression.
- Frontend: React 19, TypeScript, Vite, Tailwind CSS, TanStack Query, Recharts, Zustand.
- Data and deployment: PostgreSQL/SQLite, Docker Compose, Nginx, Render blueprint.
- Quality: pytest, Vitest, Ruff, ESLint, health checks, structured logs.

Results

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The implemented submission demonstrates a complete inference-to-action loop rather than a standalone model. Automated test suites cover system APIs, real-time behavior, forecasting, recommendations, interventions, datasets, production readiness, and frontend workflows.

Repository evidence supports the architectural and functional results; the package intentionally avoids inventing plant ROI or model-accuracy figures not present in validated artifacts.

Future Scope

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- Pilot with live PLC/SCADA integration and plant-specific calibration.
- Add authenticated roles and an identity-aware gateway.
- Move streaming fan-out and rate limits to shared infrastructure for multi-replica scale.
- Introduce governed online learning after sufficient evaluated outcomes.
- Add fleet-level benchmarking, energy optimization, and prescriptive scheduling.